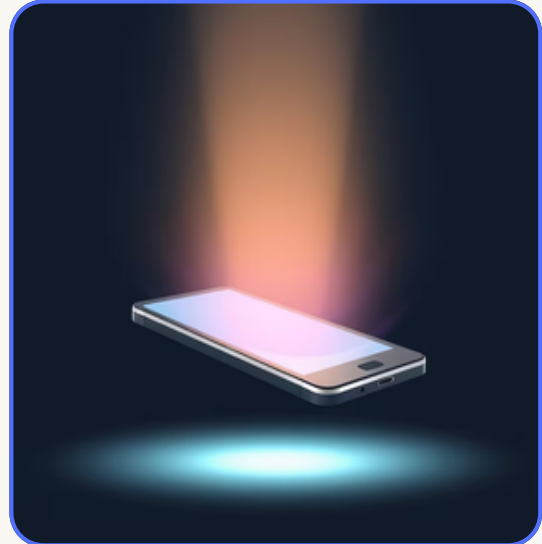


Workshop 5

THE LIGHT PHONE

EQUIPMENT LIST

- 5 high-brightness LEDs and 2 laser diodes
- 3 light-dependent resistors (LDRs)
- 3 - 3.5 mm audio jack cables
- 1 portable speaker with auxiliary input
- 5 transistors (2N2222) and 2 - 9V batteries



INSTALLATION STEPS

- Connect the smartphone to the transmitter circuit (LED) using the jack cable.
- Connect the light-dependent resistor (LDR) to the audio speaker's jack input.
- Position the LED transmitter and the LDR receiver face-to-face at a distance of 1 to 2 meters.
- Play a song with heavy bass on the smartphone.
- Check that the LED flashes in time with the sound and that the speaker is playing the music.

The Invisible Thread of Light (Wow Effect)

Ask a child to slowly move their hand between the LED and the sensor. The music stops instantly, then starts again, as soon as they remove their hand. This magically demonstrates that music “physically” travels along an invisible thread of light, one that can be intercepted !

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